

The Principia's fifth postulate: joint determinacy of place and motion

Result, 2026-09-27 (user's goal: identify the statement whose change gives mechanics with $\hbar > 0$, as the parallel postulate does for curved geometry). The statement is **joint determinacy**: the place of a body and its quantity of motion are quantities given together, so that the observables of a body multiply commutatively. Newton never states it as a postulate. He uses it in Definition II, in Lemma X and in the polygon of Proposition I, and its explicit trace is the last sentence of the scholium closing Book I, Section I: the quantities compared are “always to be diminished without limit”. Four theorems make the analogy with Euclid's fifth postulate exact.

1. **Independence (Theorem A).** For every value of a real constant $\hbar$, the Moyal product $*_{\hbar}$ on the observables of one degree of freedom satisfies Newton's second law as an identity of observables, and the flows of Newton's own integrable cases (inertia, uniform gravity, force proportional to distance) are the classical flows and automorphisms of the product. The Laws cannot decide joint determinacy, which holds exactly when $\hbar = 0$.
2. **Exactly one constant (Theorem B).** Every associative product on the observables that deforms the commutative one, is covariant under the group generated by those three flows (the affine symplectic group), and respects complex conjugation, is a Moyal product $*_{\hbar}$ with one real constant $\hbar$, which has units of action.
3. **The floor (Theorem C).** For $\hbar \neq 0$, every regular state obeys $\Delta q \Delta p \geq |\hbar|/2$, and every recorded comparison of the inertial line with the constant-force parabola obeys the floor of the Planck paper, Theorem 6: $\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{j}{2} \sum_j \Delta(\hat{X}_j) \geq |\hbar| \arcsin(1 - 2\epsilon)$. For $\hbar = 0$ there is no floor (Theorem I of the unit-and-indeterminacy note).
4. **The mechanical Wallis postulate (Theorem D).** The similarity that rescales every action, $(q, p) \mapsto (\lambda q, \lambda p)$, is a symmetry of $*_{\hbar}$ exactly when $\hbar = 0$. Wallis showed that Euclid's fifth postulate is equivalent to the existence of similar figures of different size; here joint determinacy is equivalent to mechanical similarity of recorded motions at every scale, and $\hbar$ plays the part of the absolute length of non-Euclidean geometry. Up to isomorphism there are exactly two structures, $\hbar = 0$ and $\hbar \neq 0$.

The mathematics is established: Moyal's product (Moyal 1949, metadata), Groenewold's observation that its bracket reduces to the Poisson bracket for quadratic functions (Groenewold 1946, metadata), the deformation-quantization programme (Bayen, Flato, Fronsdal, Lichnerowicz and Sternheimer 1978, metadata), and Robertson's inequality (Robertson 1929, metadata). The uniqueness statement of Theorem B is standard in substance; the proof below is self-contained. The contribution is the identification: which statement of the *Principia* is the one to change, and the proof that the Laws are independent of it in Newton's own integrable cases.

1. The textual anchor

The scholium closing Book I, Section I (1687 text, held in this repository's Latin companion), ends:

Igitur in sequentibus, siquando facili rerum imaginationi consulens, dixero quantitates quam minimas, vel evanescentes vel ultimas, cave intelligas quantitates magnitudine determinatas, sed cogita semper diminuendas sine limite.

“So in what follows, if for ease of imagination I speak of quantities as least, or vanishing, or ultimate, do not understand quantities determinate in magnitude, but think of them always as to be diminished without limit.” Newton defends this against the objection from Euclid, *Elements*

X, and applies it to the sagitta of Lemma X and the chords of Proposition I. The sentence is about geometry; its application to motion needs one more, unstated, premise: that the place of the body and its quantity of motion (Definition II) are given together at each instant, so that the sagitta measured from a place and the impulse inferred from a change of motion can both be diminished without limit in one and the same comparison. That premise is joint determinacy. Its algebraic form is the commutativity of the product of observables.

The other division scholium, after Lemma XI, says of contact angles that between any two a new series can be inserted, “Neq; novit natura limitem”. That statement concerns unobserved insertion, and it remains true for every $\hbar$: the refinement note shows that inserting unobserved times composes exactly at every $\hbar$. The change is confined to recorded comparisons.

2. The family of products

Observables of one degree of freedom (the transverse coordinate of Galileo’s comparison) are polynomials in q and p with complex coefficients. Write $\mu(f \otimes g) = fg$ and

$$P = \partial_q \otimes \partial_p - \partial_p \otimes \partial_q, \quad f *_\hbar g = \mu \circ \exp\left(\frac{i\hbar}{2}P\right)(f \otimes g),$$

a finite sum on polynomials. For $\hbar = 0$ it is the commutative product; $q *_\hbar p - p *_\hbar q = i\hbar$. The Moyal bracket is $\{f, g\}_\hbar = (f *_\hbar g - g *_\hbar f)/(i\hbar) = \frac{2}{\hbar} \mu \circ \sin(\frac{\hbar}{2}P)(f \otimes g)$, with $\{f, g\}_0$ the Poisson bracket.

3. Theorem A: independence

Theorem A. Let $H = p^2/(2M) + V(q)$ with V a polynomial and $M > 0$. For every real $\hbar$:

- (a) $\{q, H\}_\hbar = p/M$ and $\{p, H\}_\hbar = -V'(q)$: the Heisenberg equations are Newton’s second law, $M\ddot{q} = -V'(q)$, as an identity of observables;
- (b) if $\deg V \leq 2$, which covers inertia ($V = 0$), uniform gravity ($V = -Fq$) and the force proportional to distance of Book I, Proposition X ($V = kq^2/2$), then $\{f, H\}_\hbar = \{f, H\}_0$ for every f : every observable evolves by the classical flow φ_t , and each φ_t is an automorphism of $*_\hbar$.

So the commutative product ($\hbar = 0$) and every Moyal product ($\hbar \neq 0$) satisfy the Laws and reproduce Newton’s integrable motions exactly; joint determinacy holds in the first and fails in the others.

Proof. The bracket $\{f, g\}_\hbar$ is the Poisson bracket plus terms $\hbar^{2j} \mu P^{2j+1}(f \otimes g)$, $j \geq 1$, each of which puts at least three derivatives on each factor. If one factor has degree at most two they vanish. This gives (a) because q and p have degree one, and (b) because H has degree two. For (b), the flow of a quadratic Hamiltonian is an affine symplectic map, and $*_\hbar$ is covariant under affine symplectic maps: translations commute with the constant-coefficient operator P , and a linear map of determinant one leaves P invariant. $\square$

4. Theorem B: exactly one action constant

Consider products on $\mathbb{C}[q, p]$ of the form $f * g = \mu \circ B(f \otimes g)$, with $B = \sum_{\alpha, \beta} c_{\alpha\beta}(q, p) \partial^\alpha \otimes \partial^\beta$ a bidifferential operator (a finite sum on each pair of polynomials), such that

- (H1) $*$ is associative with unit 1, and $c_{00} = 1$ (it deforms the commutative product);
- (H2) $*$ is covariant under the affine symplectic group: for every translation and every linear map A of determinant one, $(f \circ A) * (g \circ A) = (f * g) \circ A$;
- (H3) $\overline{f * g} = \bar{g} * \bar{f}$.

By Theorem A(b), the flows of inertia, uniform gravity and the force proportional to distance are affine symplectic maps; free flight is the shear $(q, p) \mapsto (q + tp/M, p)$, uniform gravity adds translations, and the harmonic flow is a rotation in suitable units. The shears and rotations generate the group of linear maps of determinant one. So (H2) says that the motions Newton integrated in closed form are symmetries of the product.

Theorem B. Under (H1)–(H3), $*$ = $*_{\hbar}$ for a unique real $\hbar$. The constant $\hbar$ has the units of action, the reciprocal of the units of P .

Proof. Translation covariance makes the coefficients $c_{\alpha\beta}$ constant. Then B is a polynomial (a formal series, acting finitely on polynomials) in the components of two vectors $u = \partial^{(1)}$ and $v = \partial^{(2)}$, invariant under the simultaneous action of the group of determinant-one linear maps. By the first fundamental theorem of invariant theory for $SL(2)$, such invariants are functions of the single determinant $\det(u, v) = P$. So $B = F(P)$ with F a formal power series, $F(0) = 1$. For three factors write $x = P_{12}$, $y = P_{13}$, $z = P_{23}$ for the determinants of the pairs of derivative vectors. The three 2×2 minors of a 2×3 matrix are algebraically independent, and $(f * g) * h$, $f * (g * h)$ correspond to $F(x)F(y+z)$ and $F(z)F(x+y)$. Associativity therefore gives $F(x)F(y+z) = F(z)F(x+y)$ as formal series; setting $z = 0$ gives $F(x)F(y) = F(x+y)$, so $F(x) = e^{\kappa x}$. Exchanging the two factors reverses the sign of P and complex conjugation leaves P real, so (H3) gives $\bar{F}(x) = F(-x)$, that is $\bar{\kappa} = -\kappa$, and $\kappa = i\hbar/2$ with $\hbar$ real. Units: P carries the reciprocal of the units of qp . $\square$

Only two structures up to isomorphism. The dilation $D_{\lambda}(q, p) = (\lambda q, \lambda p)$ satisfies $P((f \circ D_{\lambda}) \otimes (g \circ D_{\lambda})) = \lambda^2(P(f \otimes g)) \circ D_{\lambda}$, so $f \mapsto f \circ D_{\lambda}$ is an isomorphism from $*_{\lambda^2\hbar}$ to $*_{\hbar}$, and $(q, p) \mapsto (q, -p)$ carries $*_{\hbar}$ to $*_{-\hbar}$ (with complex conjugation). All products with $\hbar \neq 0$ are therefore isomorphic, and distinct from the commutative one. Geometry has three constant-curvature classes (flat, spherical, hyperbolic); mechanics has two. The value of $\hbar$ in physical units is fixed by an external standard, as the absolute length of non-Euclidean geometry is.

5. Theorem C: the floor on the recorded comparison

A state is a linear functional ω on the observables with $\omega(1) = 1$ and $\omega(\bar{f} *_{\hbar} f) \geq 0$; it is regular if its GNS representation integrates to Weyl operators.

Theorem C. (a) For $\hbar \neq 0$ every state satisfies $\Delta q \Delta p \geq |\hbar|/2$. (b) For $\hbar \neq 0$ and regular states, every recorded comparison of the inertial line with the constant-force parabola, over a cell of duration τ with sagitta $s = F\tau^2/(8M)$ and impulse $J = F\tau$, made by any instrument at error probability $\epsilon < \frac{1}{2}$, leaves undetermined impulses $\hat{D}_j$ and displacements $\hat{X}_j$ with

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \geq |\hbar| \arcsin(1 - 2\epsilon) > 0.$$

(c) For $\hbar = 0$, with the instruments of Theorem I of the unit-and-indeterminacy note available, the body's posterior area can be made arbitrarily small: there is no floor.

Proof. (a) The form $(f, g) \mapsto \omega(\bar{f} *_{\hbar} g)$ is positive semidefinite; apply Cauchy–Schwarz to $f = q - \omega(q)$, $g = p - \omega(p)$ and use $q *_{\hbar} p - p *_{\hbar} q = i\hbar$ (Robertson's argument). (b) A regular state's GNS representation satisfies the Weyl relations with constant $\hbar$; by the Stone–von Neumann theorem it is a multiple of the Schrödinger representation. Theorem 6 of the Planck paper is proved there from the Weyl relations and Mandelstam–Tamm in the Bures angle, with $\hbar$ as the constant, and applies with $|\hbar|$ after the reflection $(q, p) \mapsto (q, -p)$ if $\hbar < 0$. (c) is Theorem I. $\square$

Newton's scholium asks that the compared quantities be “always diminished without limit”. Theorem A says his Laws allow it or forbid it equally; Theorem C says that in every model other than the commutative one, the recorded comparison of the inertial line with the parabola cannot be diminished below the single constant of Theorem B.

6. Theorem D: the mechanical Wallis postulate

Theorem D. For $\lambda \neq \pm 1$, the action-rescaling similarity D_λ is an automorphism of $*_{\hbar}$ if and only if $\hbar = 0$.

Proof. By §4, D_λ carries $*_{\lambda^2\hbar}$ to $*_{\hbar}$, and $*_{\lambda^2\hbar} = *_{\hbar}$ iff $\lambda^2\hbar = \hbar$. $\square$

Wallis replaced Euclid's fifth postulate by the existence of a triangle similar to a given one at any size; the non-Euclidean geometries are exactly those without similar figures of different size, and they possess an absolute unit of length. Theorem D is the mechanical counterpart. Mechanical similarity that rescales actions, the principle behind comparing motions at every scale, holds exactly when joint determinacy does; its failure is the existence of an absolute action. The dimensional note states the same fact from the other side: an admitted similarity that rescales the observable forces the floor to be zero (its Theorem B).

7. What remains, and consequence for STATE

The theorems are exact; the identification of joint determinacy as the statement to change is interpretive, because Newton used the premise without stating it. Two limits of scope. Theorem B concerns one degree of freedom and bidifferential products; several degrees of freedom follow with $Sp(2n)$ in place of $SL(2)$, whose invariants of two vectors are again generated by the symplectic pairing. And the independence of Theorem A is independence from the Laws and from Newton's integrable cases; it leaves open which physical premise, stated before quantum mechanics, forces $\hbar \neq 0$. That is STATE item 2, where Theorem U supplies the unit's value from radiation thermodynamics and Theorem I says what must be denied. This note shows that the denial of joint determinacy is, up to isomorphism, a single step with a single constant.